

## Question 3 – Bank Method Overriding

Date: 24/07/2026

### Objective / Aim

To demonstrate method overriding in Java by defining different rates of interest in RBI, SBI, ICICI, and PNB classes.

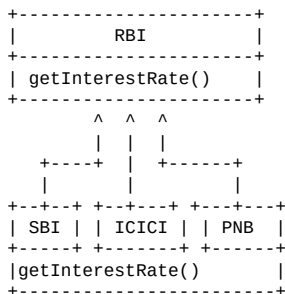
### Algorithm / Logic

1. Create the RBI superclass with getInterestRate().
2. Return 4.0 as the RBI rate.
3. Create SBI, ICICI, and PNB classes extending RBI.
4. Override getInterestRate() in each bank class.
5. Read the bank name from the user.
6. Create the selected bank object using an RBI reference.
7. Display the corresponding rate of interest.

### Steps for Execution

1. Create the Java project and add RBI, SBI, ICICI, PNB, and the main class.
2. Compile and run Week7BankMethodOverriding.java.
3. Enter RBI, SBI, ICICI, or PNB.
4. The program creates the corresponding object.
5. Observe the overridden interest rate.

### UML Class Diagram



### Source Code

```
import java.util.Scanner;

class RBI {
    double getInterestRate() { return 4.0; }
}

class SBI extends RBI {
    @Override double getInterestRate() { return 7.0; }
}

class ICICI extends RBI {
    @Override double getInterestRate() { return 6.5; }
}

class PNB extends RBI {
    @Override double getInterestRate() { return 6.0; }
}

public class Week7BankMethodOverriding {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter bank name (RBI/SBI/ICICI/PNB): ");
        String bankName = scanner.nextLine().trim().toUpperCase();
        RBI bank;
        switch (bankName) {
            case "SBI": bank = new SBI(); break;
            case "ICICI": bank = new ICICI(); break;
            case "PNB": bank = new PNB(); break;
            default: bank = new RBI(); bankName = "RBI";
        }
        System.out.printf("%s rate of interest is: %.1f%%\n", bankName, bank.getInterestRate());
        scanner.close();
    }
}
```

### Output

```
Enter bank name (RBI/SBI/ICICI/PNB): SBI
SBI rate of interest is: 7.0%
```